



Exploring the characteristics of the market-driven firms and antecedents to sustained competitive advantage

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Abstract

During the last few decades, identifying and examining the characteristics of market-driven firms have been a dominant theme in strategic marketing research. It has been argued that market-driven firms are superior in their market sensing and customer linking capabilities, enabling market-driven firms to outperform their competitors. This paper reports the findings of a study that examines the role market-focused learning capability and marketing capability in innovation-based competitive strategy on sustainable competitive advantage. The findings indicate that entrepreneurship is an important factor in sustained competitive advantage (SCA) and while market-focused learning capability leads to higher degrees of innovation, marketing capability enables SCA.

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1. Introduction

During the last four decades, the attributes of market-driven firm have been a dominant theme in strategic marketing research. The term market-driven refers to learning, understanding, and responding to stake holder perceptions and behaviors within a given market structure (Jaworski, Kohli, & Sahay, 2000). It is argued that companies that are better equipped to respond to market requirements and anticipate changing conditions enjoy long-run competitive advantage and superior profitability (Day, 1994a). Despite the growth of literature on market-driven firms, our understanding of the characteristics of such firms and their activities (Baker & Sinkula, 1999; Day, 1994a) is still limited. Researchers stress the need to explore the role of entrepreneurship and innovation to advance our knowledge of market-driven firm perfor-

mance. There is growing recognition that innovation should be incorporated into models of how organizations adapt to their environment and how this affects performance. It has been argued that as part of enhancing our understanding of market-driven firm performance, the relationship among organizational innovation, learning, and market orientation should be examined in more depth (Hurley & Hult, 1998). This paper takes this argument and expands Day's (1994a) framework, which argues that organizations can become more market oriented by identifying and building special capabilities that set them apart as market-driven organizations.

The framework presented here suggests that market-focused learning and marketing capability are critical capabilities shaping innovation-based competitive strategies in market-driven firms. Entrepreneurship plays a key role in building and nurturing market-driven capabilities. The paper contributes to the competitive strategy theory by refining and testing measures of entrepreneurship, market-driven capabilities, organizational innovation, and sustained competitive advantage (SCA), exploring the links among these constructs.

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2. Conceptual framework

The capability-based theory of SCA that has gained prominence in strategy research over the last decade suggests that a firm can achieve competitive advantage through its distinctive or core capabilities (Grant, 1991; Hayes, Pisano, & Upton, 1996; Prahalad & Hammel, 1990). Capability-based theory effectively explains the value creation process by assigning a proactive and dominant role to key decision makers of the firm. Distinctive capabilities that enable firms to gain competitive advantage are developed consciously and systematically by the willful choices and actions of the firm's strategic leaders (Child, 1972; Grant, 1991; Hayes et al., 1996; Lado, Boyd, & Wright, 1992). The conceptual framework presented in Fig. 1 is based on the theoretical propositions of the capability-based theory of competitive advantage. We argue that market-driven strategies are initiated by firm's entrepreneurial key—decision makers, their learning orientation, marketing capabilities possessed, and degree of innovativeness.

The framework presented here incorporates two market-driven capabilities, namely, market-focused learning and marketing capabilities. Market-focused learning capability represents firm's market-sensing efforts and falls within the category of outside-in capabilities in Day's (1994a) framework. The purpose of outside-in capabilities is to connect the processes that define the other organizational capabilities to the external environment and enable the business to compete by anticipating market requirements ahead of competitors and creating durable relationships with customers, channel members, and suppliers.

Marketing capability on the other hand represents the skills and resources that enable the business to add value to its products and services and falls within the category of spanning capabilities in Day's (1994a) framework. We argue that learning from markets and having the ability to reach the targeted customers with value-added products are critical capabilities for the firms pursuing innovation-based competitive strategy. Similar to Day, it is argued that market-driven firms are argued to be superior in market-focused

learning capabilities and marketing capabilities, and when these capabilities are deeply embedded within the organization, all functional activities and organizational processes will be better directed toward anticipating and responding to changing market requirements.

2.1. Entrepreneurship

In the context of market-driven competitive strategy, entrepreneurship has been identified as a significant antecedent to strategy (Hult, Thomas, & Ketchen, 2001; Miles & Arnold, 1991). Entrepreneurship is often defined in terms of the characteristics of key decision makers within the firm, particularly the personality traits and sociocultural characteristics of these decision makers (Kets de Vries, 1977; McClelland, 1961; Miller & Friesen, 1983). The entrepreneurial firm is generally distinguished in its ability to innovate, initiate change, and rapidly react to change flexibly and adroitly (Naman & Slevin, 1993). Premised on the firm behavior model of entrepreneurship (Covin & Slevin, 1986, 1991; Naman & Slevin, 1993) that has gained popularity among strategy researchers over recent years, in this paper, entrepreneurship is conceptualized as a firm behavior in which the firm displays innovativeness, proactiveness, and risk-taking propensity in their strategic decision making. It's further argued that entrepreneurship can be conceptualized in terms of a continuum using these three attributes, which reflects the "entrepreneurial intensity" of the firm. Rizzoni (1991) establishes a link between entrepreneurship and organizational capabilities, suggesting that a firm pursuing an innovation-based strategy will build specific capabilities. Such capabilities will distinguish the firm from its competitors and enable it to face the variability of the environment and outperform competitors.

2.2. Market-focused learning

Market-focused learning focuses on the capacity of the firm to learn from markets or the marketplace. To be

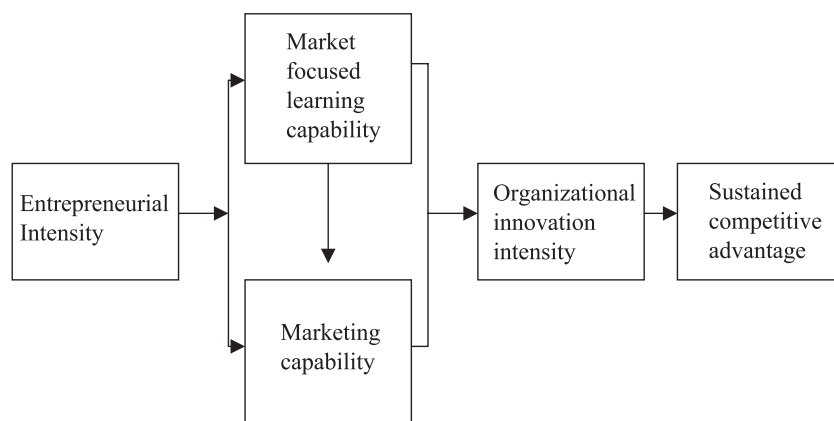


Fig. 1. A conceptual model of market-driven capabilities and organizational innovation-based competitive strategy.

effective innovators, organizations should constantly scan the marketplace for new opportunities to satisfy customers (Leavitt, 1960). This has been a thrust in market orientation research on market-driven firms (Kohli & Jaworski, 1990; Sinkula, Baker, & Noordewier, 1997). Because knowledge of market preferences (customer needs and wants) reduces the degree of incompatibility of new products with customer needs, it is likely to enhance the adoption and success of innovations (Cooper & Kleinschmidt, 1987). Based on Huber (1991), market-focused learning is defined as the capacity of the firm to acquire, disseminate knowledge, and use market information for organizational change and unlearn as and when needed.

Given that a fundamental activity of entrepreneurship is to not only create products ahead of competitors but also to create them ahead of the recognition of an explicit need by customers, market-focused learning appears an important characteristic of entrepreneurial firms. Such learning enhances a deeper understanding of the marketplace and assists in achieving this by focusing on customers' latent needs (Brown, 1991; Prahalad & Hamel, 1990), and often in minimizing the risk and maximizing learning, successful innovators frequently work with lead customers (Von, 1986). While our conceptualization of market-focused learning captures learning from both customer needs and competitor activities, we argue that market-focused learning capability is critical for entrepreneurial firms pursuing innovation-based competitive strategy and as such it is hypothesized that

Hypothesis 1: Entrepreneurial intensity has a significant effect on market-focused learning.

2.3. Marketing capability

Marketing capability is defined as integrative processes designed to apply the collective knowledge, skills, and resources of the firm to the market-related needs of the business. This enables the business to add value to its goods and services and meet competitive demands (Day, 1994a) via its marketing mix strategy. Distinctive marketing capabilities enable a firm to outperform competitors by reaching target markets effectively. Although competing firms may focus on similar market needs, the idiosyncratic way in which each firm integrates knowledge creates unique and potentially different ways of solving similar customer needs (Vorhies & Harker, 2000). Entrepreneurially oriented firms may be more inclined to implement anticipatory and responsive marketing strategies (e.g., Dess, Lumpkin, & Covin, 1997; Morris & Paul, 1987). Marketing is believed to be a means by which the firm can activate its entrepreneurial orientation or as an important complement to it (e.g., Slater & Narver, 1995). Recent research (Knight, 2000) has found that firms with a strong entrepreneurial orientation appear to be more inclined to leverage marketing strategies for entering new product markets and coping with more

complex environments. Based on the preceding discussion, it is argued that entrepreneurial firms build and nurture distinctive marketing capabilities, and as such it is hypothesized that

Hypothesis 2: Entrepreneurial intensity has a significant effect on marketing capability.

We argue that having a better understanding of the markets served by a firm enhances its capacity to undertake superior marketing programs aimed at creating added value for target markets. Organizational learning influences organizational behavior via the direct application of knowledge to solve a problem (Menon & Varadarajan, 1992). As such, market-focused learning enhances the marketing capability of the firm, and approaches to target marketing suggest that understanding the marketplace is a prerequisite for effective development and implementation of marketing mix strategies to reach the desired market segment. As such it is hypothesized that

Hypothesis 3: Market-focused learning has a significant effect on marketing capability.

2.4. Organizational innovation

In the context of innovation-based competitive strategy, the role of technological innovation and especially product innovation has often been stressed. However, the need for a broad conceptualization of innovation is well recognized (Damanpour, 1991; Hyvarinen, 1990; Porter, 1990). Building on this perspective, we define innovation as the application of ideas that are new to the firm to create added value either directly for the enterprise or indirectly for its customers, regardless of whether the newness and added value are embodied in products, processes, work organization or management, or marketing systems. In reality, what we may see is that entrepreneurial market-driven firms actively learn about changing consumer preferences and competitor behavior and actively integrate marketplace knowledge to both technological and non-technological value-creating activities of the firm. Accordingly, it is hypothesized that

Hypothesis 4: Market-focused learning has a significant effect on organizational innovation.

Marketing capability is critical at the product development stage, involving preliminary tasks such as market study, concept testing (Atuahene-Gima, 1995), and similarly a well-functioning interface between marketing and R&D is a prerequisite for successful product development (Calantone, Benedetto, & Divine, 1993), and as such it is hypothesized that

Hypothesis 5: Marketing capability has a significant effect on organizational innovation.

2.5. Sustained competitive advantage

In contemporary business, creating and maintaining a competitive advantage is becoming critical. SCA can be conceptualized as a superior “marketplace position” that captures the provision of superior customer value and/or the achievement of lower relative costs, resulting in market share dominance and superior financial performance (Hunt & Morgan, 1995). Within the broader perspective of capability-based theory, SCA is conceptualized as the firm possessing superior financial and market advantages (Day & Wensley, 1988), which are not capable of being duplicated by competitors, including the firm’s competitive strategy on which (Grant, 1991) its advantages have been founded.

It has been suggested that innovation and competitive advantage processes are closely interrelated. Porter (1990) suggests that firms create competitive advantage by conceiving new ways to conduct activities in the value chain for delivering superior value to customers, which is an act of innovation. Empirical evidence supporting the view that innovation leads to competitive advantage (stronger performance) has been indicated by Hyvarinen (1990), Lengnick-Hall (1992), and Rothwell (1992). It has been argued that innovative capabilities distinguish the more successful from the less successful growing small- and medium-size enterprises (Statistics Canada, 1994), and a recent Australian study found that firms use both technological and non-technological innovations to gain competitive advantage in both domestic and global markets (Australian Manufacturing Council [AMC], 1995). As such, it is hypothesized that

Hypothesis 6: Organizational innovation has a significant effect on SCA.

There is evidence to suggest that strong marketing capabilities are a prerequisite for the marketplace success

of innovations (Li & Calantone, 1998; Song, Montoya-Weiss, & Schmidt, 1997). Research examining the antecedents of product success and failure broadly suggests that inadequate marketing support is a key factor leading to product failure (Crawford, 1991; Davis, 1988). Adequate performance of all marketing activities is a determinant of product market success (Cooper & Kleinschmidt, 1987), and thus it is argued that marketing capability leads to SCA. Therefore, it is hypothesized that

Hypothesis 7: Marketing capabilities have a significant effect on sustained competitive advantage.

Fig. 2 provides a graphical representation and summary of the hypotheses stated above for the constructs and their relationships.

3. Research design

The study was based on the development and administration of a self-completed mail survey sent to key informants at a senior level in companies (for a similar approach, see Prasad, Ramamurthy, & Naidu, 2001). The sampling frame for the study was a list of 1272 manufacturing firms in a regional area provided by a government department. The decision to select a sample of firms from the manufacturing industry was driven by the aim of examining all types of innovation within the firm including product innovation. A further criterion, which needs to be considered, is to avoid heterogeneity of technological processes used by the firms that has implications for industry effects in research (Dess, Ireland, & Hitt, 1990). Therefore, it was decided to select two industry subgroups within the manufacturing industry, namely, the machinery and equipment manufacturing and the metal product manufacturing groups. These two industry

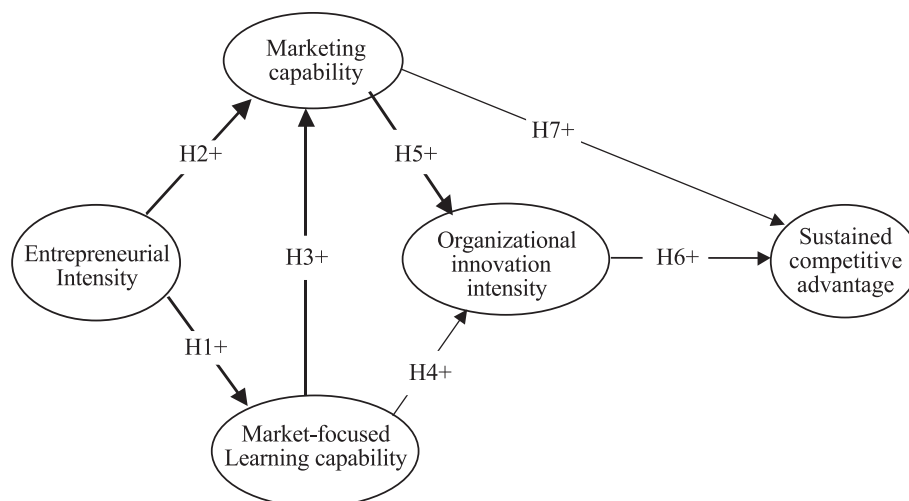


Fig. 2. Detailed conceptual model with hypothesized relationships.

subgroups are engaged in metal-based manufacturing activities and have no vast differences in their adopted technological processes.

3.1. Measures

3.1.1. Entrepreneurial intensity

The entrepreneurial intensity scale captures the extent to which the firm's strategic leaders are proactive, innovative, and risk seeking. High scores on this scale indicate that the firm's key decision makers value innovation, are proactive, and have a high tolerance for risk. The specific items used are derived from Covin and Slevin (1986) and Naman and Slevin (1993).

3.1.2. Market-focused learning capability

The market-focused learning capability scale captures the extent to which market knowledge is acquired, disseminated, unlearned, and integrated into value creation activities of the firm. High scores on the market-focused learning scale indicate that the firm possesses distinctive capabilities in the acquisition of knowledge on consumer preferences and competitor behavior in terms of the four learning activities indicated above. Firms that score highly on this scale collect market information frequently and have a thorough understanding of market preferences. The measure developed for this construct is an adaptation of the market learning scale proposed by Day (1994b).

3.1.3. Marketing capability

The marketing capability scale captures firm's customer service, advertising effectiveness, quality of sales force, strength of distribution networks, market research ability, speed of new product introduction, and ability to differentiate products. High scores on the marketing capability scale suggest that the firm possesses distinctive capabilities in the use of marketing tools and techniques. The measure used was an adaptation of the marketing capability measure originally developed by Atuahene-Gima (1993).

3.1.4. Organizational innovation intensity

The innovation intensity scale captures the extent of the firm's product, production process, managerial, and marketing innovations. This reflects a broader conceptualization of innovation that incorporates both technological and non-technological innovation. High scores on the innovation intensity scale indicate that the firm has introduced radical innovations in its product, production process, managerial, and marketing systems. This measure incorporates both the degree and types of innovation.

3.1.5. Sustained competitive advantage

The competitive advantage scale captures the extent to which the firm's innovations and distinctive capabilities resist erosion by competitors' efforts. The composite measure developed for this study is based on Day and Wensley's

(1988) view that measures of SCA should reflect more than simply financial performance. High scores on the competitive advantage scale suggest that innovations have enabled the firm to achieve superior market advantages and competitors are unable to duplicate the firm's innovations and distinctive capabilities on which advantages have been founded.

3.2. Key informants

A survey was sent to the chief executive officer of each firm following similar procedures to Prasad et al. (2001). The use of such informants has also been highlighted by Leonidou, Katsikeas, and Samiee (2002) in the meta-analysis study. The requirements for selection was that the executive was expected to (1) be conversant with the overall strategic direction of their business and (2) have direct input into the strategic decision-making process. Instructions were developed to ensure that the respondents understood the purpose of the study and that they were able to provide answers only in relation to the particular business unit involved. The key informant or alternatively self-typing approach was used in this research study. With the key informant approach, data are collected from a senior manager or a group of senior managers on information pertaining to the whole organization (or business unit). It is assumed that such senior managers have the best vantage point for viewing the entire organization and thus will provide the most accurate responses. Data on strategy gathered from middle and lower managers have questionable validity because these managers typically do not have access to information about how the total system operates (Kotha & Vadlami, 1995). Similarly, researchers have found that CEOs provide data as reliable and valid as multiple informants (Zahra & Covin, 1993). The job titles given by the respondents in the field study indicated that the sample was in fact made up of senior managers and top decision makers. As an additional measure, as suggested in Kumar, Stern, and Anderson (1993), a self-assessment of knowledgeable ability was adopted and the mean score for manager knowledge in the study was greater than 5, indicating evidence of knowledgeable ability and suitability of the informants.

It is important to note that although in marketing management the central concern has been with the development of the marketing mix and in the management/policy area with higher level issues associated with the corporate mission, competitive advantage, and grand strategies, the pivotal element in each is the manager or decision maker. Whether the process emphasized is synoptic or incremental (Lyonski & Pecotich, 1992) or the focus on strategy is situation specific, universal, or contingency, it is the firms management who on the basis of perceptions manages the alignment of environmental opportunities and threats, internal firm strengths and weaknesses, and strategic action taken (Aaker, 2001; Cravens, 1999; Jain, 1999). A strong

case can therefore be made that in the context of innovation, learning and marketing strategic actions are more likely to be consistent with management perceptions than with objective criteria (Miles, Snow, & Pfeffer, 1974). In reality, strategic action takes place in a social context and while objective conditions, and events may influence management actions, it is perceptions associated with these conditions that are of great importance. The view that management perception should be the focus of research in marketing has been advocated, particularly top level management (Wind & Robertson, 1983).

3.3. Administration

In total, 326 useable questionnaires that were returned yielded a response rate of 25.6%. This response rate is within the range of response rates of recent field survey research in marketing (e.g., Lusch & Brown, 1996).

The sampling method succeeded in providing observations that varied greatly, at least in terms of firm characteristics. The firms represented in the sample varied in size, as measured by annual sales (mean=\$14 million; S.D.=\$44 million), number of employees (mean=65 employees; S.D.=177 employees). On average, the firms in the sample had been operating for around 24 years (S.D.=21.7 years). Approximately 58% of firms in the sample were competing in export markets for 9 years on average (S.D.=8.1 years).

Missing values were treated using listwise deletion, which resulted in an effective sample size of 324 observations. Although the loss of observations can be large in some situations, this method produces consistent estimates of model parameters (Bollen, 1989). In total, however, only two observations were deleted (less than 1%) using this method. Nonresponse bias was assessed using an extrapolation method that compared waves of early and late respondents (Armstrong & Overton, 1977). This procedure showed no significant differences between early and late respondents, which suggests that nonresponse bias may not be a problem.

3.4. Analytic technique

After data screening, confirmatory factor analysis for each construct was computed. Following this, the reliability, convergent, and discriminant validity of measures were assessed. Next, composite variables were created using the factor score regression weights. Finally, the structural model was estimated to test the hypotheses.

4. Results

Preliminary data analysis was undertaken to examine the psychometric properties of the scales via measures of central tendency, dispersion, bivariate Pearson correlations, reliability estimates, and confirmatory factor analysis. To evaluate

the fit of the CFA's and SEM's, several goodness-of-fit indicators were used, including (using χ^2 statistic) goodness-of-fit index (GFI), adjusted goodness-of-fit index (AGFI), root mean square residual (RMSR), and root mean square error of approximation (RMSEA) (O'Cass, 2000). The preliminary results presented in Table 1 indicate that the measures were acceptable. Table 1 also provides confirmatory factor analysis results for the measurement models for each construct. The CFA results indicate that all constructs met acceptable benchmarks on the key fit indices. The preliminary results indicated that acceptable reliability and levels of fit on the indices used to judge the confirmed factor models exists. These preliminary results indicated that the constructs proposed were being tapped by the measurement instrument developed. The results of the preliminary analyses of the instrument indicate acceptance of the psychometric properties of the scales, providing ample support to move to the next stage of the study involving the full scale estimation and testing of the hypotheses.

As single sources of information can introduce spurious relationships among the variables, and as this study collected data via the same method (self-report scales), the need to test for common method variance was warranted. This test was conducted in accordance with Harmon's one factor test where all items, presumably measuring a variety of different constructs, were subjected to a single factor analysis. Using this approach, eight factors were extracted with eigenvalues greater than 1 and the variance explained was 62.4%. As there was not one factor (or a common factor underlying the data) and as the majority of the variance was not accounted for by one general factor, a substantial amount of common method variance was not evident.

Convergent validity tests indicated that all variances extracted were greater than 0.50, indicating convergent validity (Fornell & Larcker, 1981). Discriminant validity tests indicated that all reliabilities for constructs were greater than their respective correlations and as such discriminant validity was found (Gaski, 1984).

Table 1
Preliminary results

Variable	Means	S.D.	(1)	(2)	(3)	(4)
(1) SCA	2.984	0.898				
(2) Innovation intensity	3.097	0.825	.496			
(3) Market-focused learning	3.532	0.946	.368	.414		
(4) Entrepreneurial intensity	3.348	0.804	.494	.581	.439	
(5) Marketing capability	3.460	0.700	.331	.345	.616	.391

Measurement model results	χ^2	RMSR	GFI	AGFI	RMSEA	α
(1) SCA	3.84	0.024	.99	.99	0.001	.80
(2) Innovation intensity	33.25	0.060	.99	.98	0.05	.86
(3) Market-focused learning	7.42	0.02	.99	.98	0.038	.83
(4) Entrepreneurial intensity	12.51	0.03	.99	.98	0.03	.83
(5) Marketing capability	21.16	0.05	.98	.97	0.03	.83

* All correlations significant at $P < .01$. All P values for the measurement models were nonsignificant.

To test the hypotheses, the relationships depicted in the hypotheses and shown in Fig. 2 were estimated using structural equation modeling (SEM) via Lisrel. Estimating the structural model produced a nonsignificant chi-square statistic ($\chi^2=2.322$, $P=.313$). Other fit indices too provided evidence of adequate fit to the sample data (GFI=.997, AGFI=.974, normed fit index=.997, nonnormed fit index=.998). The structural model explains 19.3%, 49.8%, 41.4%, and 37.5%, respectively, of the variation in market-focused learning capability, marketing capability, organizational innovation intensity, and SCA providing additional support for the structural model.

Standardized parameter estimates and t values for the model are shown in Table 2, and the results support all hypotheses (H1–H4, H6 and H7, except for H5). The results indicate that both market-focused learning capability and marketing capability are influenced by entrepreneurial intensity, supporting H1 and H2. Firms that are innovative, proactive, and having a risk-seeking posture are more likely to possess distinctive market-focused learning capabilities ($\beta=.385$, $t=7.713$) and distinctive marketing capabilities ($\beta=.506$, $t=3.176$).

As predicted by H3, there is a strong positive relationship between market-focused learning capability and marketing capability ($\beta=.825$, $t=12.388$). Entrepreneurial firms that are excellent in market learning capabilities are more likely to possess excellent marketing capabilities. Results support H4, indicating a positive relationship between market-focused learning capability and organizational innovation intensity ($\beta=.206$, $t=3.480$). Entrepreneurial firms, which actively learn from markets, are more likely to market innovative products and services and introduce new and value-adding changes to production processes, marketing methods, and managerial systems. Results do not support H5, indicating that marketing capability does not lead to innovation intensity.

The results indicate that there is a positive relationship between organizational innovation intensity and SCA, supporting H6. Organizations are more likely to sustain market advantages over their competitors where they have success-

fully implemented radical changes to their ways of doing business ($\beta=.374$, $t=4.684$). Finally, the model suggests that marketing capability is positively related to SCA ($\beta=.111$, $t=2.201$). In addition to the relationships predicted in the initial model, the results also suggest that entrepreneurial firms achieve higher innovation intensity ($\beta=.548$, $t=8.870$) and SCA ($\beta=.504$, $t=3.074$), strengthening the emerging marketing/entrepreneurship interface paradigm that entrepreneurial intensity is a key factor in the market-driven competitive strategy. Our discussion attempts to shed some light on the path that firms must follow to achieve this position.

5. Discussion and directions for future research

Collectively, the results provide strong support for the conceptual framework. The theoretical constructs operate largely as hypothesized and explain a substantial proportion of the variation in SCA. A central theme of our research is that entrepreneurship is a key factor in the market-driven competitive strategy. The conceptual model suggests that entrepreneurial firms pursuing innovation-based competitive strategy build and nurture distinctive capabilities in market-focused learning and marketing. The results of the study largely support this position. These relationships have not been examined in prior research.

Market-focused learning capability was operationalized in terms of the key organizational learning processes suggested in the literature (Huber, 1991; Sinkula, 1994), namely, knowledge acquisition, knowledge dissemination within the firm, integration of waves of knowledge acquired into organizational innovation, and actively undertaking unlearning. The confirmatory factor model supports all these activities except for information dissemination reflecting that sampled firms do not engage in explicit knowledge sharing activities. This may be partly due to the fact that the sample is dominated by a large number of small firms. Small firms are closely controlled by owner/managers who make key managerial decisions and lack functional levels that impede the flow of information. Therefore, there is no need for multifunctional activities and discussions and conflict resolution procedures (Cooper & Kleinschmidt, 1991; Cosier & Schwenk, 1990), which may be essential for larger and decentralized firms.

Entrepreneurial intensity is an important determinant of the market-focused learning capability and marketing capability, providing support to the capability theory of SCA. Although the capability-based theory assigns a dominant role to firm's entrepreneurial key decision makers, this relationship has not been examined in prior research. As predicted, market-focused learning leads to organizational innovation supporting the view that markets are a key source of idea generation in the innovation process. Entrepreneurial firms having distinctive market-focused learning capabilities integrate the waves of knowledge acquired from markets into firm's value-creating activities.

Table 2
Standardized parameter estimates for hypothesized model

Path	Parameter estimate	t Value
H1: Entrepreneurial intensity $\rightarrow$ market-focused learning	0.385	7.713
H2: Entrepreneurial intensity $\rightarrow$ marketing capability	0.506	3.176
H3: Market-focused learning $\rightarrow$ marketing capability	0.825	12.388
H4: Market-focused learning $\rightarrow$ innovation intensity	0.206	3.480
H5: Marketing capability $\rightarrow$ innovation intensity	ns	ns
H6: Innovation intensity $\rightarrow$ SCA	0.374	4.684
H7: Marketing capability $\rightarrow$ SCA	0.111	2.201

The findings relating to the marketing capability contribute to our understanding on the role of marketing in the strategy dialogue (Day, 1992; Kerin, 1992). The findings suggest that entrepreneurship is a key factor driving the marketing capability building process. Similarly, distinctive market-focused learning leads to distinctive marketing capabilities. Firm's superior knowledge about market changes and market segments enable the firm to effectively use key marketing tools to reach target markets. Although marketing capability does not lead to innovation intensity, it leads to SCA reflecting that superior skills in handling marketing mix elements, speed of new product introduction, and ability to differentiate products facilitate market success of innovations. Overall, findings of the study support the view that "the functional role of marketing in strategic management deals with the entrepreneurial work of the organization and organizational renewal and growth; in short, innovation" (Kerin, 1992, p. 332).

The findings suggest that entrepreneurial firms pursue organizational innovation. Prior research suggests that the components of the entrepreneurship scale used in this study are positively related to technological innovation (Covin & Slevin, 1986). In the current study, innovation construct incorporates both technological and nontechnological innovation. The findings therefore suggest that entrepreneurial organizations pursue both technological and nontechnological innovation and both types of innovation lead to SCA.

The study examined only two types of market-driven capabilities suggested in Day's (1994a) framework. Future research may examine the role of inside-out capabilities, which is the third category of capabilities that are activated by market requirements, competitive challenges, and external opportunities. Similarly, few firms possess all the market-driven capabilities required to achieve a competitive advantage. Arguably, developing distinctive capabilities involves a trade-off in terms of which capabilities to develop. Further research might be usefully directed towards exploring how these decisions are made. A growing number of researchers argue that a firm cannot be "world class" in everything it does and that some of the capabilities can be outsourced. Similarly, future research might explore other antecedents of marketing capabilities, one of which is physical resources owned by the firm (Grant, 1991).

6. Managerial implications

Arguably, the framework used in this paper presents managers with a feasible path for building SCA. Firms aspiring to achieve competitive advantage must adopt an entrepreneurial posture in their key decisions, and such firms build and nurture market-focused learning capabilities. These capabilities in turn enable them to develop superior marketing capabilities. The tacitness, complexity, and specificity of the capabilities possessed by the firm can produce causal ambiguity regarding the firm's actions, which in turn

create barriers to imitation. However, managers should recognize that building distinctive capabilities could be an expensive undertaking (Teece, Pisano, & Shuen, 1990). The process involves first developing certain capabilities and then selecting and refining a few of those capabilities to the point where they become the basis for competitive advantage. Hayes et al. (1996, p. 505) summarize the process stating that

Converting an ability to a real capability does not happen by accident: it takes time (not just man-hours, elapsed time) and consumes resources. And building such a capability into something that can become the basis of a competitive advantage requires a dedication that approaches fanaticism; the time and resources required often well exceed what one can logically justify.

The manner in which organizational innovation intensity is conceptualized has important implications for practitioners and policymakers. In a clear departure from the existing literature, which is biased towards product-based technological innovation, the model implies that innovation can occur in any of the firm's value creation activities. All forms of innovation are potential sources of SCA. This approach encourages firms to explore all paths to innovation including not only new products, but also new marketing methods, employee reward schemes, and new organizational structures. Policymakers must take cognizance of the importance of both technological and nontechnological innovations in achieving competitive advantage. Accordingly, they must formulate policies encouraging firms to adopt a market-focused approach in their competitive capability building activities.

Similarly, the manner in which entrepreneurship is conceptualized has implications for practitioners. The model presented here implies that any firm can adopt an entrepreneurial posture by displaying innovativeness, a high tolerance for risk taking, and proactiveness in their business decisions. In our theoretical framework, entrepreneurs are not "born" as suggested in conventional theories, rather firms can be guided and educated to become entrepreneurial in their decision making through strategic leadership. This interpretation also has important implications for policy planning. For example, managers can be educated and trained to engage in firm-level entrepreneurship.

7. Limitations

Though the study provides some useful insights into the role of market-driven capabilities in innovation-based competitive strategy, certain limitations should be recognized. First, the study used single-informant reports to measure each of the theoretical constructs. An alternative approach would have been to combine information from multiple informants, although the practical difficulties associated

with using information from multiple-informant reports are well recognized by management researchers (Kumar et al., 1993). Second, the cross-sectional research design limits the extent to which inferences can be made about the causal ordering of variables. Finally, the manufacturing sectors that we studied provided an appropriate setting, although research in other industries is required. Consider that the pace of technological change in some industries may differ from the manufacturing sectors that we studied.

8. Conclusions

Although the literature on the role of market-driven capabilities in competitive strategy has grown in significance over the last decade, few efforts have been made to operationalize the key constructs and synthesize the central propositions in a single framework. This study contributes to the capability-based theory of competitive advantage by developing measures for two key market-driven capabilities, namely, market-focused learning and marketing, and examining their role in the innovation-based competitive strategy. Further, the model captures the critical role of key decision makers in the development of market-driven capabilities. The study also contributes to our understanding on the role of marketing in the strategy dialogue. For practitioners, the results of the study provide a feasible path for developing competitive advantage. Focusing exclusively on technological advances ignores the many potential sources of non-technological innovation that managers can vigorously pursue to achieve SCA. In addition to its empirical contribution, it is hoped that this research will focus the attention of researchers and managers on the crucial role that strategic leadership plays in developing market-driven capabilities and shaping the firm's competitive position and SCA.

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